Project Report: Customer Segmentation & Advanced Churn Analysis

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Role: Business Analyst Intern

Tools Used: Python, Pandas, NumPy, Matplotlib

Domain: Telecom Customer Analytics

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1. Project Overview

This project focuses on analyzing telecom customer data to understand **churn behavior** and identify customer segments that are at higher risk of leaving the service. Using segmentation and exploratory analysis, the project reveals insights that can help businesses build better **retention strategies** and improve customer satisfaction.

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2. Business Objective

The primary objective of this project was to answer a key business question: **"Which customer segments are most likely to churn, and what factors contribute to their behavior?"**

Telecom companies lose significant revenue due to high churn rates. By identifying patterns and segmenting customers, we aim to :

- Reduce churn
- Improve customer experience
- Increase customer loyalty
- Support business decision-making with data-backed insights

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3. Dataset Description

The dataset contains telecom customer records with features such as:

- **Demographics:** Gender, SeniorCitizen, Dependents
- **Services:** InternetService, TechSupport, Streaming
- **Account Information:** Contract, PaymentMethod, MonthlyCharges, TotalCharges
- **Tenure:** Number of months with the company

- **Churn Label:** Whether the customer left or stayed

Rows: **7,043** Columns: **21**

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4. Data Cleaning & Preprocessing

To ensure accurate analysis, the dataset was cleaned using Python:

✓ Missing Values Handled

- “TotalCharges” contained blank strings → converted to numeric
- Missing values imputed using **median/mode** where necessary

✓ Incorrect Data Types Fixed

- Converted TotalCharges → float
- Ensured numeric columns were in correct format

✓ Standardized “Churn” Column
The Churn column was cleaned to ensure consistency by:

- Removing extra spaces
- Converting all values to lowercase
- Standardizing values into categorical labels "Yes" and "No"

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5. Exploratory Data Analysis (EDA)

5.1 Visualizations Used

- **Donut Chart:** Customer Segmentation by Tenure
- **Bar Chart:** Average Monthly Charges per Segment
- **Churn Rate Comparison:** Based on gender, senior citizens, contract type, payment method

5.2 Key Observations

✓ Customer Segmentation Share
- 50.6% → Loyal customers (37+ months)
- 29.5% → Mid-tenure customers
- 20.9% → New customers

✓ Average Monthly Charges by Segment

- 0–12 months: **$61**
- 13–36 months: **$66.5**
- 37+ months: **$49.1**

✓ Churn Behavior Patterns

- **Month-to-month contract** customers churn the most
- **Higher monthly charges** strongly correlate with higher churn
- **Senior citizens** show a higher churn probability
- **Electronic check** payment method is associated with higher churn
- **Long-tenure customers (37+)** are less likely to churn

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6. Conclusion

The project successfully highlights the **key factors contributing to churn** and identifies which customer groups require immediate attention. The combination of segmentation, visualization, and churn analysis provides a clear picture of customer behavior—helping businesses make **data-driven retention strategies**.

This project strengthened my skills in:

- Business Analysis
- Data Cleaning & Preprocessing
- Customer Segmentation
- Insight Generation
- Visualization & Storytelling